

Name: .....



**Gosford High School**

**Trial HSC 2022**

## Mathematics Extension 2

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### General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Section II, show relevant mathematical reasoning and/ or calculations

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### Total Marks

**100**

#### Section I – 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section
- Answer questions on the Multiple-choice answer sheet

#### Section II – 90 marks

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section

# Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the best response on the multiple-choice answer sheet.

1. Given  $z = \frac{1 + \sqrt{3}i}{1 + i}$ , the modulus and argument of the complex number  $z^5$  are respectively

(A)  $2\sqrt{2}, \frac{6}{5\pi}$

(B)  $4\sqrt{2}, \frac{12}{5\pi}$

(C)  $4\sqrt{2}, \frac{12}{7\pi}$

(D)  $2\sqrt{2}, -\frac{12}{\pi}$

2. Which vector represents the unit vector in the direction  $\begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix}$ ?

(A)  $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

(B)  $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

(C)  $\frac{7}{1} \begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix}$

(D)  $\frac{1}{49} \begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix}$

3. The angle between the vectors  $\tilde{u} = 3\tilde{i} + 5\tilde{j} + 7\tilde{k}$  and  $\tilde{v} = 11\tilde{i} + 13\tilde{j} + 17\tilde{k}$  is closest to

(A)  $8^\circ$

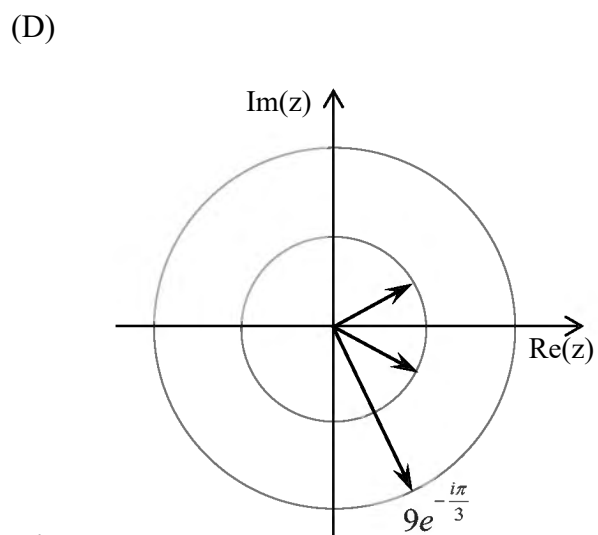
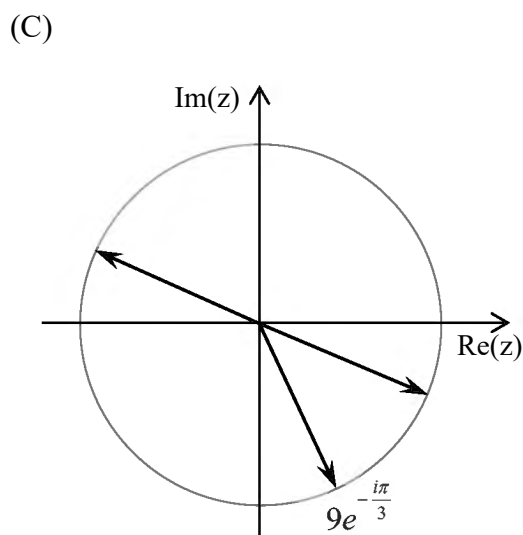
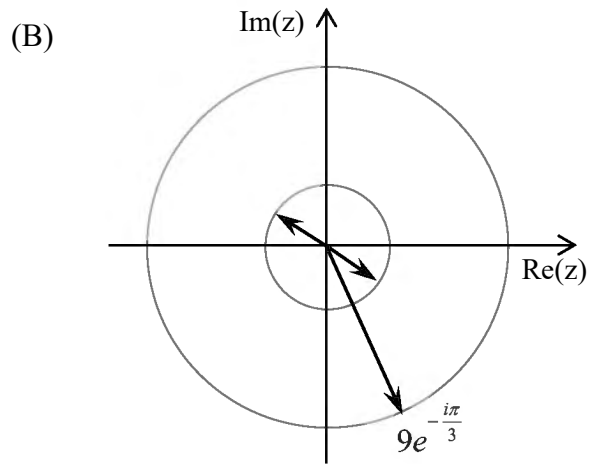
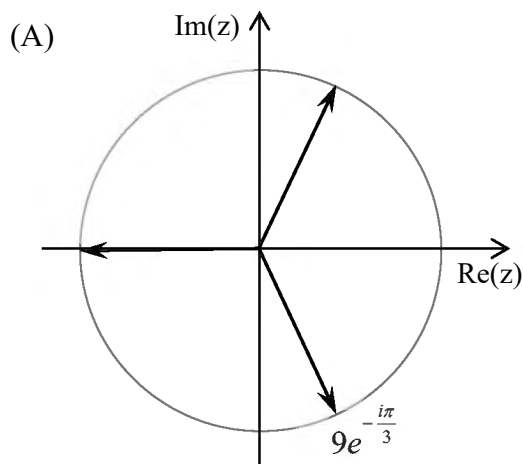
(B)  $11^\circ$

(C)  $79^\circ$

(D)  $82^\circ$

4.  $(\sqrt{3} + i)^n$  is purely real when
- (A)  $n = 6k, k \in \mathbb{Z}$
- (B)  $n = 6k + \pi, k \in \mathbb{Z}$
- (C)  $n = \frac{k}{6}, k \in \mathbb{Z}$
- (D)  $n = \frac{6}{k}, k \in \mathbb{Z}$
5. Which of the following is the contrapositive of the statement “If Star Wars was released then I did not go school” ?
- (A) If I went to school then Star Wars was released.
- (B) If I did not go to school then Star Wars was not released.
- (C) If Star Wars was not released then I went to school.
- (D) If I went to school then Star Wars was not released.

6. The diagram which best represents the two square roots of  $9e^{\frac{i\pi}{3}}$  is:



7. Which of the following statements is NOT always true.
- (A)  $|a+b| \geq |a|-|b|$
- (B)  $|a-b| \leq |a|-|b|$
- (C)  $|a+b| \leq |a|+|b|$
- (D)  $|a-b| \leq |a|+|b|$
8. The polynomial  $P(z) = z^4 - 4z^3 + Mz + 20$ ,  $M \in \mathbb{R}$ . Given that  $3+i$  is a zero of  $P(z)$ , which of the following could be a quadratic factor of  $P(z)$ ?
- (A)  $z^2 + 2z + 2$
- (B)  $z^2 - 6z + 4$
- (C)  $z^2 + 6z + 10$
- (D)  $z^2 - 2z + 20$
9. Which of the following relations has a graph that passes through  $1+2i$  on the complex plane?
- (A)  $\text{Arg}(z) = \frac{\pi}{6}$
- (B)  $|z-1| = |z-2i|$
- (C)  $\text{Re}(z) = 2\text{Im}(z)$
- (D)  $z + \bar{z} = 2$
10. Evaluate  $\int_0^1 \tan^{-1} x \, dx$
- (A)  $\frac{\pi}{4} - \ln \sqrt{2}$
- (B)  $\frac{\pi}{6} - \ln \sqrt{3}$
- (C)  $\frac{\pi}{2} - \ln \sqrt{5}$
- (D)  $\frac{\pi}{2} - \ln \sqrt{6}$

## Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours 45 minutes for this section

Answer in the appropriate booklet provided

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**Question 11.** (15 marks) Answer in the booklet labelled *Question 11*

a) Given that  $z_1 = 2 - i$ ,  $z_2 = 1 + 5i$ , express the following in the form  $a + bi$ , where  $a, b$ , are real.

(i)  $(\overline{z_1})$  1

(ii)  $\frac{z_1}{z_2}$  2

(iii)  $|z_2 - z_1|$  2

(iv)  $z_1 \overline{z_1}$  2

b) Prove by counter-example that the proposition  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x = y^2$  is false. 1

c) Find the cube roots of  $8i$  and clearly represent the solutions on an Argand diagram. 3

d) Let  $\overrightarrow{OA} = -\underline{\hat{i}} + 4\underline{\hat{k}}$  and  $\overrightarrow{OB} = 4\underline{\hat{i}} + \underline{\hat{j}} + 3\underline{\hat{k}}$ .

(i) Find  $\overrightarrow{AB}$ . 1

(ii) Find  $|\overrightarrow{AB}|$  1

(iii) Find the coordinates of point C such that  $\overrightarrow{AC} = -\underline{\hat{i}} + \underline{\hat{j}} - \underline{\hat{k}}$  2

**Question 12** (14 Marks) Answer in the booklet labelled *Question 12*

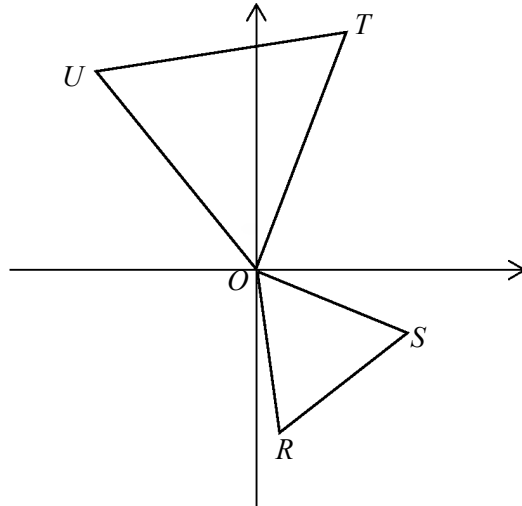
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- a) (i) Show that  $\int \frac{2 + \ln x}{x^2} dx$  can be expressed as  $\int (2 + u)e^{-u} du$  **2**
- (ii) Hence, or otherwise, determine the exact value of  $\int_1^e \frac{2 + \ln x}{x^2} dx$  **3**
- b) Determine the vector equation  $\vec{r}$  of the line segment  $AB$  where  $A = (1, 2, 1)$  and  $B = (4, 10, -3)$ . **2**
- c) (i) Express  $\frac{2x}{(x+1)(x^2+1)}$  in the form  $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$  where A, B and C are real. **3**
- (ii) Hence or otherwise, find  $\int_0^1 \frac{2x}{(x+1)(x^2+1)} dx$  **4**

- a) Prove by contradiction that for positive real numbers  $a$ ,  $b$ , and  $c$ , if  $ab = c$  then  $a \leq \sqrt{c}$  or  $b \leq \sqrt{c}$  2
- b) Given that  $3 + qi$  is one of the roots of the quadratic equation  $z^2 + 12pz + 58 = 0$ , where  $p$  and  $q$  are real constants, find the values of  $p$  and  $q$ . 3
- c) A sequence is defined by the recursive formula  $T_{n+1} = \frac{1}{5}(T_n^2 + 6)$ ,  $T_1 = \frac{5}{2}$ . Prove  $T_{n+1} < T_n \quad \forall n > 0, n \in \mathbb{Z}$  by the process of mathematical induction. 3
- d) Find the cartesian equation of the curve given by  $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + \frac{1}{t+1}\mathbf{j}$  and hence find the maximum value of the function. 3
- e) Represent on the Argand diagram where the two inequalities  $|z - i| \leq 2$  and  $0 \leq \arg(z + 1) \leq \frac{\pi}{4}$  both hold. 3

- a) The diagram shows points  $O, R, S, T$  and  $U$  in the complex plane. These points correspond to the complex numbers  $0, r, s, t$  and  $u$  respectively.  $\triangle ORS$  and  $\triangle OTU$  are equilateral.

Let  $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$



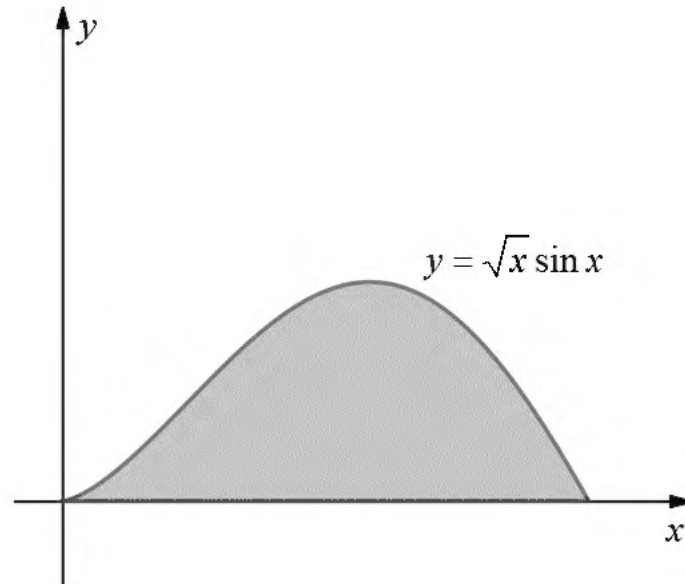
- (i) Explain why  $u = zt$  1
- (ii) Express the complex number  $r$  in terms of  $s$ . 2
- (iii) Use complex numbers to show that the lengths of  $RT$  and  $SU$  are equal. 3
- b) Prove by mathematical induction that  $\sum_{k=0}^n \frac{1}{3^k} = \frac{3}{2} \left( 1 - \frac{1}{3^{n+1}} \right)$  for positive integers  $n$ . 3
- c) It is known that  $\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$ . DO NOT PROVE THIS.
- If  $x, y$  and  $z$  are the interior angles of a triangle, show that  $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{\pi}$  3
- d) Determine the Cartesian equation of the sphere with centre  $(9, 20, 12)$  and passes through the point  $(25, 0, 0)$ . 3



a) (i) Find  $\int \sin^2 x \, dx$  2

(ii) Show that  $\int x \sin^2 x \, dx = \frac{1}{8}(2x^2 - 2x \sin 2x - \cos 2x) + C$  3

(iii) The diagram shows a finite region bound by the curve  $y = \sqrt{x} \sin x$  and the  $x$  axis.



Find the volume of revolution when the shaded region is rotated about the  $x$  axis. 3

b) Let the roots of  $z^3 = -4 + 4\sqrt{3}i$  be  $z_1, z_2$  and  $z_3$ .

(i) Find  $z_1, z_2$  and  $z_3$ , expressing your answers in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . 3

(ii) Hence, or otherwise, show that  $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) = 0$  2

c) Let  $y = x^2 e^x$ . Prove by mathematical induction that

$$\frac{d^n}{dx^n} x^2 e^x = (n(n-1) + 2nx + x^2) e^x \quad \forall n \in \mathbb{Z}^+. \quad 3$$

- a) By using the substitution  $t = \tan \frac{x}{2}$  or otherwise, find  $\int \frac{1}{2 + \sin x} dx$  **3**
- b) The integral  $I_n$  is defined by  $I_n = \int_0^1 e^x (2-x)^n dx$  for integers  $n \geq 0$ .
- (i) Show that  $I_n = e - 2^n + nI_{n-1}$  **2**
- (ii) Hence or otherwise, evaluate  $\int_0^1 e^x (2-x)^4 dx$  **3**
- c) (i) Show that both  $\sin^2 x + \cos^4 x$  and  $\sin^4 x + \cos^2 x$  can each be expressed as  $1 - \frac{1}{4} \sin^2 2x$  **2**
- (ii) Show that  $\frac{1}{\sin^2 x + \cos^4 x} + \frac{1}{\sin^4 x + \cos^2 x} = \frac{8}{4 - \sin^2 2x}$  **1**
- (ii) Hence, or otherwise, show that  $2 \leq \frac{1}{\sin^2 x + \cos^4 x} + \frac{1}{\sin^4 x + \cos^2 x} \leq \frac{8}{3}$  **2**
- d) Prove that  $a^3 + 3a^2 + 2a$ , where  $a$  is a positive whole number, is divisible by 6 by mathematical induction. **3**

**End of Exam**



**Gosford High School**

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Mathematics Extension 2

**Sample Solutions and Marking Guidelines**

## Section I

|    |   |
|----|---|
| 1  | B |
| 2  | C |
| 3  | A |
| 4  | A |
| 5  | D |
| 6  | B |
| 7  | B |
| 8  | A |
| 9  | D |
| 10 | A |

## Section II

### Question 11. (14 marks)

a) Given that  $z_1 = 2 - i$ ,  $z_2 = 1 + 5i$ , express the following in the form  $a + bi$ , where  $a, b$ , are real.

(i)  $(\bar{z}_1)$

1

|                     |                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 mark              | Correct answer                                                                                                                                                         |
| Markers<br>Comments | a) Students performed very well in parts (i), (ii), (iii) and (iv), however attention to detail would have eliminated some basic errors when manipulating expressions. |

### Sample solution

$$\bar{z}_1 = 2 + i$$

(ii)  $\frac{z_1}{z_2}$

2

|                     |                                                  |
|---------------------|--------------------------------------------------|
| 2 marks             | Correct answer                                   |
| 1 mark              | Attempt to realise the denominator or equivalent |
| Markers<br>Comments | See above                                        |

### Sample solution

$$\begin{aligned}
 \frac{z_1}{z_2} &= \frac{2-i}{1+5i} \\
 &= \frac{2-i}{1+5i} \times \frac{1-5i}{1-5i} \\
 &= \frac{2-10i-i+5i^2}{1-25i^2} \\
 &= \frac{-3-11i}{26} \\
 &= -\frac{1}{26}(3+11i)
 \end{aligned}$$

(iii)  $|z_2 - z_1|$

2

|                 |                                                                           |
|-----------------|---------------------------------------------------------------------------|
| <b>2 marks</b>  | Correct answer                                                            |
| <b>1 mark</b>   | Evaluates $z_2 - z_1$ correctly or determines modulus of a complex number |
| <b>Markers</b>  |                                                                           |
| <b>Comments</b> | See above                                                                 |

**Sample solution**

$$\begin{aligned}
 |z_2 - z_1| &= |(1 + 5i) - (2 - i)| \\
 &= |-1 + 6i| \\
 &= \sqrt{(-1)^2 + (6)^2} \\
 &= \sqrt{37}
 \end{aligned}$$

(iv)  $z_1 \bar{z}_1$

2

|                 |                                 |
|-----------------|---------------------------------|
| <b>2 marks</b>  | Correct answer                  |
| <b>1 mark</b>   | Progress towards correct answer |
| <b>Markers</b>  |                                 |
| <b>Comments</b> |                                 |

**Sample solution**

$$\begin{aligned}
 z_1 \bar{z}_1 &= |z_1|^2 \\
 &= 2^2 + (-1)^2 \\
 &= 5
 \end{aligned}$$

b) Prove by counter-example that the proposition  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x = y^2$  is false.

1

|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| <b>1 mark</b>   | Provides a valid counter-example                                                  |
| <b>Markers</b>  |                                                                                   |
| <b>Comments</b> | a) Any real negative value for $x$ would have been sufficient to gain the 1 mark. |

**Sample solution**

Let  $x = -1$ . There is no real  $y$  such that  $y^2 = -1$ . Hence the proposition is false.

c) Find the cube roots of  $8i$  and clearly represent the solutions on an Argand diagram.

3

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3 marks</b>          | Obtains all three correct solutions with representation on the Argand diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>2 marks</b>          | Obtains all three cube roots or represents solutions on the Argand diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>1 mark</b>           | Obtains one correct cube root or equivalent merit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Markers Comments</b> | <p>a) Generally well done, however students who adopted the following approach will encounter difficulties when dealing with higher powers of <math>z</math>.</p> $z^3 = 8i \Rightarrow z^3 + 8i^3 = 0, \quad \therefore (z + 2i)(z^2 - 2iz + 4i^2) = 0$ <p><i>i.e.</i> <math>(z + 2i)(z^2 - 2iz - 4) = 0</math></p> <p>Hence <math>z = 2i</math> or <math>z = \frac{2i \pm \sqrt{-4 + 16}}{2}</math>, leading to <math>z = -2i</math> or <math>z = i \pm \sqrt{3}</math>.</p> <p>Students are reminded to show the roots on a circle of radius 2</p> |

### Sample solution

$$\text{Let } z^3 = 8i$$

$$= 2^3 \operatorname{cis}\left(\frac{\pi}{2}\right)$$

$$z_1 = 2 \operatorname{cis}\left(\frac{1}{3} \cdot \frac{\pi}{2}\right)$$

$$= 2 \operatorname{cis}\left(\frac{\pi}{6}\right)$$

$$z_2 = 2 \operatorname{cis}\left(\frac{\pi}{6} + \left(\frac{2\pi}{3}\right)\right)$$

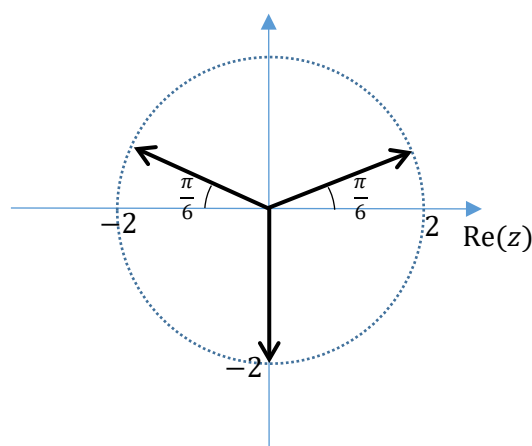
$$= 2 \operatorname{cis}\left(\frac{5\pi}{6}\right)$$

$$z_3 = 2 \operatorname{cis}\left(\frac{\pi}{6} + 2\left(\frac{2\pi}{3}\right)\right)$$

$$= 2 \operatorname{cis}\left(\frac{9\pi}{6}\right)$$

$$= 2 \operatorname{cis}\left(-\frac{\pi}{2}\right)$$

$$\therefore z_1 = \sqrt{3} + i, z_2 = -\sqrt{3} + i, z_3 = -2i$$



d) Let  $\overrightarrow{OA} = -\underline{i} + 4\underline{k}$  and  $\overrightarrow{OB} = 4\underline{i} + \underline{j} + 3\underline{k}$ .

(i) Find  $\overrightarrow{AB}$ .

1

|          |                                                      |
|----------|------------------------------------------------------|
| 1 mark   | Correct answer                                       |
| Markers  |                                                      |
| Comments | Most students performed well on this vector question |

### Sample solution

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix}$$

$$= 5\underline{i} + \underline{j} - \underline{k}$$

(ii) Find  $|\overrightarrow{AB}|$

1

|          |                |
|----------|----------------|
| 1 mark   | Correct answer |
| Markers  |                |
| Comments |                |

### Sample solution

$$|\overrightarrow{AB}| = \sqrt{5^2 + 1^2 + (-1)^2}$$

$$= \sqrt{27}$$

$$= 3\sqrt{3}$$



(iii) Find the coordinates of point C such that  $\overrightarrow{AC} = -\underline{i} + \underline{j} - \underline{k}$

1

|          |                |
|----------|----------------|
| 1 mark   | Correct answer |
| Markers  |                |
| Comments |                |

### Sample solution

$$\begin{aligned}\overrightarrow{OC} &= \overrightarrow{OA} + \overrightarrow{AC} \\ &= \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}\end{aligned}$$

$$\therefore C(-2, 1, 3)$$

**Question 12** (14 Marks)

- a) (i) Show that  $\int \frac{2 + \ln x}{x^2} dx$  can be expressed as  $\int (2 + u)e^{-u} du$

**2**

|                         |                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2 marks</b>          | Result logically shown                                                                                                                                                                                                                       |
| <b>1 mark</b>           | Identifies appropriate substitution and correct derivative.                                                                                                                                                                                  |
| <b>Markers Comments</b> | a) (i) Show that $\int \frac{2 + \ln x}{x^2} dx$ , required student to show the result step by step rather than jump steps to reach $\int (2 + u)e^{-u} du$ . A few students had difficulty in finding the appropriate expression for $dx$ . |

**Sample solution**

Let  $u = \ln x$ , hence  $x = e^u$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$\begin{aligned} \int \frac{2 + \ln x}{x^2} dx &= \int (2 + \ln x) \frac{1}{x} \cdot \frac{1}{x} dx \\ &= \int (2 + u) \frac{1}{e^u} du \\ &= \int (2 + u)e^{-u} du \end{aligned}$$

- (ii) Hence, or otherwise, determine the exact value of  $\int_1^e \frac{2 + \ln x}{x^2} dx$

**3**

|                         |                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------|
| <b>3 marks</b>          | Correct solution                                                                      |
| <b>2 marks</b>          | Evaluation of new limits and application of integration by parts, or equivalent merit |
| <b>1 mark</b>           | Evaluation of new limits or application of integration by parts, or equivalent merit  |
| <b>Markers Comments</b> | Well done, however, some students failed to change limits.                            |

**Sample solution**

$$\int_1^e \frac{2 + \ln x}{x^2} dx = \int_0^1 (2 + u)e^{-u} du$$

$$\begin{aligned} w &= 2 + u & v &= -e^{-u} \\ w' &= 1 & v' &= e^{-u} \end{aligned}$$

$$\begin{aligned} &= \left[ -e^{-u}(2 + u) \right]_0^1 - \int_0^1 -e^{-u}(1) du \\ &= \left( -\frac{2+1}{e} \right) - \left( -\frac{2+0}{1} \right) - \left[ e^{-u} \right]_0^1 \\ &= -\frac{3}{e} + 2 - \left( \frac{1}{e} - 1 \right) \\ &= 3 - \frac{4}{e} \end{aligned}$$

- b) Determine the vector equation  $\vec{r}$  of the line segment  $AB$  where  $A = (1, 2, 1)$  and  $B = (4, 10, -3)$ .

2

|                         |                                                                                                                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2 marks</b>          | Correct answer, including the restrictions on $\lambda$ .                                                                                                                                                                                       |
| <b>1 mark</b>           | Identifies vector equation of the line passing through AB or equivalent                                                                                                                                                                         |
| <b>Markers Comments</b> | <p>a) Only a few students gained full marks.<br/>Many stated that <math>\lambda = 1</math>, the required result needed <math>0 \leq \lambda \leq 1</math>.<br/>Evaluating the values for <math>A, B</math> and <math>C</math> was well done</p> |

### Sample solution

$$\begin{aligned}\vec{r} &= \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4-1 \\ 10-2 \\ -3-1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix}\end{aligned}$$

But  $\vec{r}$  represents any point on the line segment  $AB$ . When  $\lambda = 0$ ,  $\vec{r} = \overrightarrow{OA}$ . When  $\lambda = 1$ ,  $\vec{r} = \overrightarrow{OB}$

$$\therefore \vec{r} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix} \text{ where } 0 \leq \lambda \leq 1$$

- c) (i) Express  $\frac{2x}{(x+1)(x^2+1)}$  in the form  $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$  where A, B and C are real.

3

|                  |                                                                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 marks          | Correct answer                                                                                                                                                          |
| 2 marks          | Correct set of simultaneous equations or equivalent                                                                                                                     |
| 1 mark           | Correct expression involving A, B and C                                                                                                                                 |
| Markers Comments | a) Most students recognised the integrals leading to log and inverse tan however some failed to separate $\frac{x+1}{x^2+1}$ into $\frac{x}{x^2+1} + \frac{1}{x^2+1}$ . |

### Sample solution

$$\frac{2x}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$$\therefore A(x^2+1) + (Bx+C)(x+1) = 2x$$

Equating coefficients:

$$x^2 : A+B=0 \quad [1]$$

$$x : B+C=2 \quad [2]$$

$$c : A+C=0 \quad [3]$$

$$\therefore A=-1, B=1, C=1$$

$$\therefore \frac{2x}{(x+1)(x^2+1)} = -\frac{1}{x+1} + \frac{x+1}{x^2+1}$$

(ii) Hence or otherwise, find  $\int_0^1 \frac{2x}{(x+1)(x^2+1)} dx$

4

|          |                                      |                                                   |
|----------|--------------------------------------|---------------------------------------------------|
| 4 marks  | Correct answer including evaluation  |                                                   |
| 3 marks  | Three correct integration components | As per the third line of the suggested solutions. |
| 2 marks  | Two correct integration components   |                                                   |
| 1 mark   | One correct integration              |                                                   |
| Markers  |                                      |                                                   |
| Comments |                                      |                                                   |

### Sample solution

$$\begin{aligned}
 \int_0^1 \frac{2x}{(x+1)(x^2+1)} dx &= \int_0^1 -\frac{1}{x+1} + \frac{x+1}{x^2+1} dx \\
 &= \int_0^1 -\frac{1}{x+1} + \frac{1}{2} \cdot \frac{2x}{x^2+1} + \frac{1}{x^2+1} dx \\
 &= \left[ -\ln|x+1| + \frac{1}{2} \ln|x^2+1| + \tan^{-1} x \right]_0^1 \\
 &= \left[ \ln \left| \frac{\sqrt{x^2+1}}{x+1} \right| + \tan^{-1} x \right]_0^1 \\
 &= -\frac{1}{2} \ln 2 + \frac{\pi}{4} - \ln 1 - 0 \\
 &= \frac{\pi}{4} - \frac{\ln 2}{2}
 \end{aligned}$$

**Question 13** (14 Marks) Answer in the booklet labelled *Question 13*

- a) Prove by contradiction that for positive real numbers  $a$ ,  $b$ , and  $c$ , if  $ab = c$  then  $a \leq \sqrt{c}$  or  $b \leq \sqrt{c}$

2

|                         |                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2 marks</b>          | Valid proof by contradiction                                                                                                                                                                                                                                                        |
| <b>1 mark</b>           | Valid negation statement                                                                                                                                                                                                                                                            |
| <b>Markers Comments</b> | Proof by contradiction requires students to firstly negate the statement. Students experienced difficulty negating and “or” statement (see solution). Once you have negated the statement you show there is a contradiction between what is shown and what you are told to be true. |

**Sample solution**

Negation statement: For positive real numbers  $a$ ,  $b$ , and  $c$ , if  $ab = c$  then  $a > \sqrt{c}$  and  $b > \sqrt{c}$ .

If  $a > \sqrt{c}$  then

$$a\sqrt{c} > c \text{ since } \sqrt{c} > 0$$

$$\Rightarrow a\sqrt{c} > ab \text{ since } ab = c$$

$$\Rightarrow \sqrt{c} > b$$

Which contradicts the condition of the negation that  $b > \sqrt{c}$

Hence the original statement must be true.

- b) Given that  $3 + qi$  is one of the roots of the quadratic equation  $z^2 + 12pz + 58 = 0$ , where  $p$  and  $q$  are real constants, find the values of  $p$  and  $q$ .

3

|                         |                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>3 marks</b>          | Correct answer                                                                                                           |
| <b>2 marks</b>          | Progress towards correct answer                                                                                          |
| <b>1 mark</b>           | Recognition that the roots occur in conjugate pairs or equivalent                                                        |
| <b>Markers Comments</b> | Students should recognise that if a polynomial with integer coefficients then complex roots occur in complex conjugates. |

**Sample solution**

$3 + qi$  is a root  $\Rightarrow 3 - qi$  must also be a root since the coefficients are all real.

$$\text{Sum of roots: } (3 + qi) + (3 - qi) = -12p$$

$$6 = -12p$$

$$p = -\frac{1}{2}$$

$$\text{Product of roots: } (3 + qi)(3 - qi) = 58$$

$$9 + q^2 = 58$$

$$q^2 = 49$$

$$q = \pm 7$$

- c) A sequence is defined by the recursive formula  $T_{n+1} = \frac{1}{5}(T_n^2 + 6)$ ,  $T_1 = \frac{5}{2}$ .

Prove  $T_{n+1} < T_n \quad \forall n > 0, n \in \mathbb{Z}$  by the process of mathematical induction.

3

|                  |                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 marks          | Valid proof by mathematical induction                                                                                                                                           |
| 2 marks          | Significant progress towards valid proof                                                                                                                                        |
| 1 mark           | Verification of the case for $n = 1$                                                                                                                                            |
| Markers Comments | Mostly well done. However some students are not writing “by the assumption” when using the induction hypothesis.<br>Other students had trouble applying the recurrence formula. |

### Sample solution

$$\begin{aligned} \text{For } n=1, \quad T_1 &= \frac{5}{2}, \quad T_2 = \frac{1}{5} \left( \left( \frac{5}{2} \right)^2 + 6 \right) \\ &= \frac{49}{20} \\ &< T_1 \end{aligned}$$

Assume true for  $n = k$ . I.e.  $T_{k+1} < T_k \quad \forall k > 0, k \in \mathbb{Z}$

R.T.P true for  $n = k + 1$ . I.e.  $T_{k+2} < T_{k+1} \quad \forall k > 0, k \in \mathbb{Z}$ . Hence, r.t.p.  $T_{k+2} - T_{k+1} < 0$

$$\begin{aligned} LHS &= T_{k+2} - T_{k+1} \\ &= \frac{1}{5} \left( (T_{k+1})^2 + 6 \right) - \frac{1}{5} \left( (T_k)^2 + 6 \right) \\ &\leq \frac{1}{5} \left( (T_k)^2 + 6 \right) - \frac{1}{5} \left( (T_k)^2 + 6 \right) \text{ from assumption} \\ &\leq 0 \text{ as required} \end{aligned}$$

Hence true by the process of mathematical induction.

- d) Find the cartesian equation of the curve given by  $z(t) = \sqrt{t}i + \frac{1}{t+1}j$  and hence find the maximum value of the function.

3

|                     |                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| 3 marks             | Cartesian equation for the function, the restriction and the maximum value.                                    |
| 2 marks             | Cartesian equation for the function and either the restriction on domain or the maximum value of the function. |
| 1 mark              | Cartesian equation for the function                                                                            |
| Markers<br>Comments | Mostly well done, although students should be mindful of the domain.                                           |

### Sample solution

$$x = \sqrt{t} \Rightarrow t = x^2 \text{ for } x \geq 0$$

$$y = \frac{1}{t+1}$$

$$\therefore y = \frac{1}{x^2+1} \text{ for } x \geq 0$$

The maximum value of  $y = \frac{1}{x^2+1}$  occurs at the same  $x$  ordinate as the minimum value of  $y = x^2 + 1$ .

Hence  $x = 0$  gives the maximum and minimum values of these respective functions.

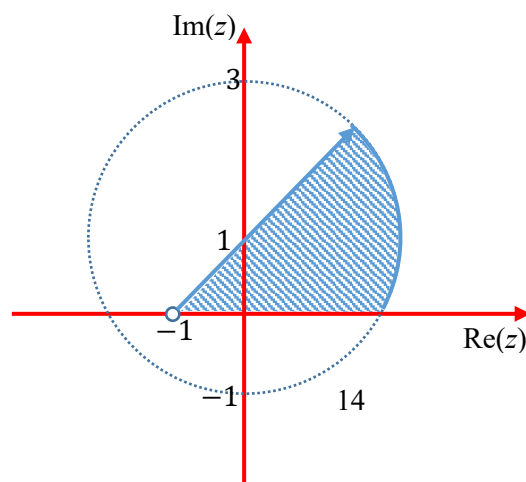
$$\text{Maximum value of } y = \frac{1}{x^2+1} \text{ is } y = \frac{1}{0^2+1} = 1$$

- e) Represent on the Argand diagram where the two inequalities  $|z-i| \leq 2$  and  $0 \leq \arg(z+1) \leq \frac{\pi}{4}$  both hold.

3

|                     |                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 3 marks             | Correct diagram                                                                                                                 |
| 2 marks             | Significant progress towards correct diagram                                                                                    |
| 1 mark              | One of the inequalities correctly represented on the diagram                                                                    |
| Markers<br>Comments | Mostly well done. Please do not draw a solid circle as part of your solution. A solid line means this is part of your solution. |

### Sample solution

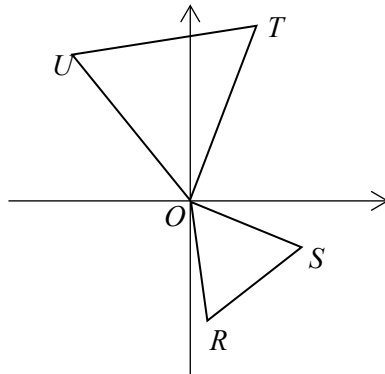




**Question 14** (14 Marks)

- a) The diagram shows points  $O, R, S, T$  and  $U$  in the complex plane. These points correspond to the complex numbers  $0, r, s, t$  and  $u$  respectively.  $\triangle ORS$  and  $\triangle OTU$  are equilateral.

Let  $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$



- (i) Explain why  $u = zt$

1

|          |                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1 mark   | Correct answer                                                                                                                    |
| Markers  | Mostly well done, although many students neglected to mention the modulus of $z$ when explaining how complex numbers are rotated. |
| Comments |                                                                                                                                   |

**Sample solution**

$|u| = |t|$  as they form sides of equilateral  $\triangle UOT$  and angle  $\angle UOT = \frac{\pi}{3}$  (angle in an equilateral triangle).

$$|z| = 1 \text{ and } \text{Arg}(z) = \frac{\pi}{3}$$

$$\begin{aligned} \text{Hence } zt &= |z||t| \text{cis}(\text{Arg}(z) + \text{Arg}(t)) \\ &= |t| \text{cis}\left(\text{Arg}(t) + \frac{\pi}{3}\right) \\ &= u \end{aligned}$$

As this gives a rotation anticlockwise by  $\frac{\pi}{3}$  from  $t$  to  $u$ .

(ii) Express the complex number  $r$  in terms of  $s$ .

1

|                     |                                                                |
|---------------------|----------------------------------------------------------------|
| 1 mark              | Correct answer                                                 |
| Markers<br>Comments | Mostly well done with some careless errors from some students. |

### Sample solution

$$r = \frac{s}{z} \text{ or } r = s\bar{z}$$

(iii) Use complex numbers to show that the lengths of  $RT$  and  $SU$  are equal.

3

|                     |                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 marks             | Correct solution                                                                                                                                |
| 2 marks             | Valid expressions for both $ \overrightarrow{SU} $ and $ \overrightarrow{RT} $ or equivalent merit                                              |
| 1 mark              | Valid expression for either $ \overrightarrow{SU} $ or $ \overrightarrow{RT} $ or equivalent merit                                              |
| Markers<br>Comments | Many different solutions to this question and students should be mindful about how they communicate their solution. Neatness and diagrams help. |

### Sample solution

$$\begin{aligned} |\overrightarrow{SU}| &= |\overrightarrow{SO}| + |\overrightarrow{OU}| \\ &= -\underline{s} + \underline{t}\underline{z} \end{aligned}$$

$$\begin{aligned} |\overrightarrow{RT}| &= |\overrightarrow{RO}| + |\overrightarrow{OT}| \\ &= -\frac{s}{z} + t \\ &= \frac{1}{z}(-\underline{s} + \underline{t}\underline{z}) \\ &= \bar{z}(-\underline{s} + \underline{t}\underline{z}) \\ &= \bar{z}|\overrightarrow{SU}| \end{aligned}$$

$$\text{But } |\bar{z}| = 1 \text{ hence } |\overrightarrow{RT}| = |\overrightarrow{SU}|$$

- b) Prove by mathematical induction that  $\sum_{k=0}^n \frac{1}{3^k} = \frac{3}{2} \left( 1 - \frac{1}{3^{n+1}} \right)$  for positive integers  $n$ .

3

|                         |                                                                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3 marks</b>          | Valid proof by mathematical induction                                                                                                                                                       |
| <b>2 marks</b>          | Significant progress towards valid proof                                                                                                                                                    |
| <b>1 mark</b>           | Verification of the case for $n = 1$                                                                                                                                                        |
| <b>Markers Comments</b> | Many students should revise summation induction proofs. Careless errors were made in showing the base case and students need to write “by the assumption” at the induction hypothesis step. |

### Sample solution

For  $n = 1$ : 
$$LHS = \frac{1}{3^0} + \frac{1}{3^1} = \frac{4}{3}$$

$$RHS = \frac{3}{2} \left( 1 - \frac{1}{3^2} \right) = \frac{4}{3}$$
Hence true for  $n = 1$

Assume true for  $n = p$ . 
$$\sum_{k=0}^p \frac{1}{3^k} = \frac{3}{2} \left( 1 - \frac{1}{3^{p+1}} \right)$$
I.e. 
$$\frac{1}{3^0} + \frac{1}{3^1} + \cdots + \frac{1}{3^p} = \frac{3}{2} \left( 1 - \frac{1}{3^{p+1}} \right)$$

Required to prove true for  $n = p + 1$ . I.e. r.t.p 
$$\frac{1}{3^0} + \frac{1}{3^1} + \cdots + \frac{1}{3^p} + \frac{1}{3^{p+1}} = \frac{3}{2} \left( 1 - \frac{1}{3^{p+2}} \right)$$

$$\begin{aligned}
 LHS &= \frac{1}{3^0} + \frac{1}{3^1} + \cdots + \frac{1}{3^p} + \frac{1}{3^{p+1}} \\
 &= \frac{3}{2} \left( 1 - \frac{1}{3^{p+1}} \right) + \frac{1}{3^{p+1}} \text{ from assumption} \\
 &= \frac{3}{2} \left( 1 - \frac{1}{3^{p+1}} + \frac{2}{3} \cdot \frac{1}{3^{p+1}} \right) \\
 &= \frac{3}{2} \left( 1 - \frac{3}{3 \cdot 3^{p+1}} + \frac{2}{3 \cdot 3^{p+1}} \right) \\
 &= \frac{3}{2} \left( 1 - \frac{1}{3^{p+2}} \right) \\
 &= RHS \text{ as required}
 \end{aligned}$$

Hence true by the process of mathematical induction.

- c) It is known that  $\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$ . DO NOT PROVE THIS.

If  $x$ ,  $y$  and  $z$  are the interior angles of a triangle, show that  $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{\pi}$

3

|                  |                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 marks          | Valid proof                                                                                                                                                                   |
| 2 marks          | Application of given inequality to the terms $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ and incorporation of $x + y + z = \pi$ or equivalent merit.                            |
| 1 mark           | Application of given inequality to the terms $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ or incorporation of $x + y + z = \pi$ or equivalent merit.                             |
| Markers Comments | Students over complicated this question by squaring $x + y + z$ . There were a few different methods for this question and students should be mindful to communicate clearly. |

### Sample solution

$$x + y + z = \pi \text{ (Interior angle sum of a triangle) } \quad \boxed{1} \quad \text{Given } \frac{x+y+z}{3} \geq \sqrt[3]{xyz} \quad \boxed{2}$$

Substituting into the given inequality  $\boxed{2}$ :

$$\frac{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}{3} \geq \sqrt[3]{\frac{1}{x} \cdot \frac{1}{y} \cdot \frac{1}{z}}$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{3}{\sqrt[3]{xyz}} \quad \boxed{3}$$

Consider the original inequality  $\boxed{2}$ :

$$\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$$

$$\frac{x+y+z}{9} \geq \frac{\sqrt[3]{xyz}}{3}$$

$$\frac{3}{\sqrt[3]{xyz}} \geq \frac{9}{x+y+z}$$

Hence  $\frac{3}{\sqrt[3]{xyz}} \geq \frac{9}{\pi}$  (from  $\boxed{1}$ )  $\boxed{4}$

Now applying result  $\boxed{4}$  to  $\boxed{3}$ :

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{\pi} \text{ as required.}$$

- d) Determine the Cartesian equation of the sphere with centre  $(9, 20, 12)$  and passes through the point  $(25, 0, 0)$ . **3**

|                             |                                         |
|-----------------------------|-----------------------------------------|
| <b>3 marks</b>              | Correct answer in the stated form       |
| <b>2 marks</b>              | Further progress towards correct answer |
| <b>1 mark</b>               | Calculation of radius                   |
| <b>Markers<br/>Comments</b> | Very well done.                         |

**Sample solution**

$$\begin{aligned}\text{radius} &= \sqrt{(25-9)^2 + (0-20)^2 + (0-12)^2} \\ &= \sqrt{800}\end{aligned}$$

$$\therefore (x-9)^2 + (y-20)^2 + (z-12)^2 = 800$$

**Question 15** (15 Marks)a) (i) Find  $\int \sin^2 x \, dx$ **1**

|                 |                |
|-----------------|----------------|
| <b>1 mark</b>   | Correct answer |
| <b>Markers</b>  | Very well done |
| <b>Comments</b> |                |

**Sample solution**

$$\begin{aligned} \cos 2x &= 1 - 2\sin^2 x & \int \sin^2 x \, dx &= \frac{1}{2} \int (1 - \cos 2x) \, dx \\ \sin^2 x &= \frac{1}{2}(1 - \cos 2x) & &= \frac{1}{2} \left( x - \frac{1}{2} \sin 2x \right) + C \end{aligned}$$

(ii) Show that  $\int x \sin^2 x \, dx = \frac{1}{8}(2x^2 - 2x \sin 2x - \cos 2x) + C$ **3**

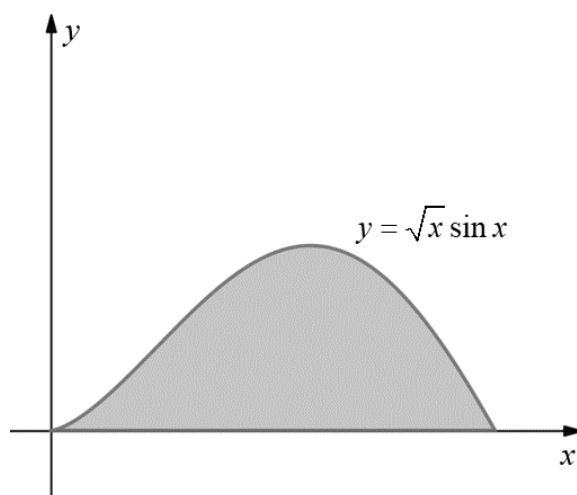
|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| <b>3 marks</b>  | Correct answer                                                           |
| <b>2 marks</b>  | Significant progress towards correct answer                              |
| <b>1 mark</b>   | Valid substitution into integration by parts formula or equivalent merit |
| <b>Markers</b>  | Very well done                                                           |
| <b>Comments</b> |                                                                          |

**Sample solution**Let  $u = x$   $v = \frac{1}{2} \left( x - \frac{1}{2} \sin 2x \right)$  from (i)

$$u' = 1 \quad v' = \sin^2 x$$

$$\begin{aligned} \int x \sin^2 x \, dx &= uv - \int v u' \, dx \\ &= \frac{1}{2} x \left( x - \frac{1}{2} \sin 2x \right) - \frac{1}{2} \int \left( x - \frac{1}{2} \sin 2x \right) \, dx \\ &= \frac{x^2}{2} - \frac{x \sin 2x}{4} - \frac{1}{2} \left( \frac{x^2}{2} + \frac{1}{4} \cos 2x \right) + C \\ &= \frac{x^2}{2} - \frac{x \sin 2x}{4} - \frac{x^2}{4} - \frac{\cos 2x}{8} + C \\ &= \frac{1}{8} (4x^2 - 2x \sin 2x - 2x^2 - \cos 2x) + C \\ &= \frac{1}{8} (2x^2 - 2x \sin 2x - \cos 2x) + C \end{aligned}$$

- (iii) The diagram shows a finite region bound by the curve  $y = \sqrt{x} \sin x$  and the  $x$  axis.



Find the volume of revolution when the shaded region is rotated about the  $x$  axis.

3

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| 3 marks             | Correct answer                                           |
| 2 marks             | Further progress towards correct answer                  |
| 1 mark              | Calculation of bounds of integration or equivalent merit |
| Markers<br>Comments | Very well done                                           |

### Sample solution

$$x \text{ intercepts: } \sqrt{x} \sin x = 0$$

$$x = 0, \pi$$

Volume of revolution:

$$\begin{aligned}
 V &= \pi \int_0^{\pi} x \sin^2 x \, dx \\
 &= \frac{\pi}{8} \left[ 2x^2 - 2x \sin 2x - \cos 2x \right]_0^{\pi} \\
 &= \frac{\pi}{8} \left[ \left( 2\pi^2 - 2\pi \sin 2\pi - \cos 2\pi \right) - \left( 2(0)^2 - 2(0) \sin 2(0) - \cos 2(0) \right) \right] \\
 &= \frac{\pi}{8} \left[ 2\pi^2 - 0 - 1 - 0 + 0 + 1 \right] \\
 &= \frac{\pi^3}{4}
 \end{aligned}$$

b) Let the roots of  $z^3 = -4 + 4\sqrt{3}i$  be  $z_1, z_2$  and  $z_3$ .

- (i) Find  $z_1, z_2$  and  $z_3$ , expressing your answers in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ .

3

|                  |                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 3 marks          | Correct answers expressed in the required form                                                                                           |
| 2 marks          | Finds one root and expresses in the form $re^{i\theta}$ or finds three roots but not expressed in the required form, or equivalent merit |
| 1 mark           | Express $z^3$ in the form $re^{i\theta}$ or equivalent merit                                                                             |
| Markers Comments | Mostly well done, some students found the modulus incorrectly                                                                            |

### Sample solution

$$z^3 = -4 + 4\sqrt{3}i$$

$$= 8e^{\frac{2\pi}{3}i}$$

Let  $z = re^{i\theta}$

$$z_1 = 2e^{\frac{2\pi}{9}i}, \quad z_2 = 2e^{\left(\frac{2\pi}{9} + \frac{2\pi}{3}\right)i}, \quad z_3 = 2e^{\left(\frac{2\pi}{9} + \frac{4\pi}{3}\right)i}$$

$$z_1 = 2e^{\frac{2\pi}{9}i}, \quad z_2 = 2e^{\left(\frac{8\pi}{9}\right)i}, \quad z_3 = 2e^{\left(-\frac{4\pi}{9}\right)i}$$

- (ii) Hence, or otherwise, show that  $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) = 0$

2

|                  |                                 |
|------------------|---------------------------------|
| 2 marks          | Result shown                    |
| 1 mark           | Valid progress towards solution |
| Markers Comments | Very well done                  |

### Sample solution

Consider the equation:

$$z^3 = -4 + 4\sqrt{3}i$$

$$z^3 + 4 - 4\sqrt{3}i = 0$$

$$\text{I.e. } z^3 + 0z^2 + 0z + 4 - 4\sqrt{3}i = 0$$

Sum of roots:  $z_1 + z_2 + z_3 = -\frac{b}{a}$

From (i):  $2e^{\frac{2\pi}{9}i} + 2e^{\left(\frac{8\pi}{9}\right)i} + 2e^{\left(-\frac{4\pi}{9}\right)i} = 0$

$$= 0 \quad e^{\frac{2\pi}{9}i} + e^{\left(\frac{8\pi}{9}\right)i} + e^{\left(-\frac{4\pi}{9}\right)i} = 0$$

Hence both the real parts and imaginary parts of the LHS must be 0.



Real part:  $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) + \cos\left(-\frac{4\pi}{9}\right) = 0$  But  $\cos(-\theta) = \cos(\theta)$

$$\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) = 0$$

$$\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) = 0 \text{ as required.}$$

c) Let  $y = x^2 e^x$ . Prove by mathematical induction that

$$\frac{d^n}{dx^n} x^2 e^x = (n(n-1) + 2nx + x^2) e^x \quad \forall n \in \mathbb{Z}^+.$$

3

|                         |                                                                               |
|-------------------------|-------------------------------------------------------------------------------|
| <b>3 marks</b>          | Valid proof by mathematical induction                                         |
| <b>2 marks</b>          | Significant progress towards valid proof                                      |
| <b>1 mark</b>           | Verification of the case for $n = 1$                                          |
| <b>Markers Comments</b> | Mostly well done, some students chose an incorrect value for the initial case |

### Sample solution

For  $n = 1$ :  $LHS = \frac{d}{dx} [x^2 e^x]$   $RHS = (1(1-1) + 2(1)x + x^2) e^x$

$$= x^2 e^x + 2x e^x \quad = e^x (x^2 + 2x)$$

$$= e^x (x^2 + 2x) \quad = LHS \text{ hence true for } n = 1.$$

Assume true for  $n = k$ :  $\frac{d^k}{dx^k} [x^2 e^x] = (k(k-1) + 2kx + x^2) e^x \quad \forall k \in \mathbb{Z}^+$

R.T.P true for  $n = k + 1$  i.e. r.t.p  $\frac{d^{k+1}}{dx^{k+1}} [x^2 e^x] = ((k+1)(k) + 2(k+1)x + x^2) e^x \quad \forall k \in \mathbb{Z}^+$

$$LHS = \frac{d^{k+1}}{dx^{k+1}} [x^2 e^x]$$

$$= \frac{d}{dx} \left[ \frac{d^k}{dx^k} [x^2 e^x] \right]$$

$$= \frac{d}{dx} [(k(k-1) + 2kx + x^2) e^x] \text{ from assumption}$$

$$= (k^2 - k + 2kx + x^2) e^x + (2k + 2x) e^x \text{ by product rule}$$

$$= (k^2 - k + 2kx + 2x + 2k + x^2) e^x$$

$$= (k^2 + k + 2kx + 2x + x^2) e^x$$

$$= ((k+1)(k) + 2(k+1)x + x^2) e^x$$

$$= RHS \text{ as required}$$

Hence true by the process of mathematical induction.

**Question 16** (16 Marks)

- a) By using the substitution  $t = \tan \frac{x}{2}$  or otherwise, find  $\int \frac{1}{2 + \sin x} dx$

**3**

|                         |                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------|
| <b>3 marks</b>          | Correct answer in terms of $x$                                                                   |
| <b>2 marks</b>          | Valid integration in terms of $t$ or equivalent merit                                            |
| <b>1 mark</b>           | Valid substitution into integral or equivalent merit                                             |
| <b>Markers Comments</b> | Well done, however, some students were unable to successfully show that $dx = \frac{2dt}{1+t^2}$ |

**Sample solution**

$$\text{Let } t = \tan \frac{x}{2} \quad \Rightarrow \sin x = \frac{2t}{1+t^2}$$

$$\begin{aligned} \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1}{2} \left( 1 + \tan^2 \frac{x}{2} \right) \\ &= \frac{1+t^2}{2} \end{aligned}$$

$$dx = \frac{2}{1+t^2} dt$$

$$\begin{aligned} \int \left( \frac{1}{2 + \sin x} \right) dx &= \int \frac{1}{2 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= \int \frac{2}{2 + 2t^2 + 2t} dt \\ &= \int \frac{1}{1 + t^2 + t} dt \\ &= \int \frac{1}{\frac{3}{4} + \left( t + \frac{1}{2} \right)^2} dt \\ &= \int \frac{1}{\left( \frac{\sqrt{3}}{2} \right)^2 + \left( t + \frac{1}{2} \right)^2} dt \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2t+1}{\sqrt{3}} \right) + C \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2 \tan \frac{x}{2} + 1}{\sqrt{3}} \right) + C \end{aligned}$$

b) The integral  $I_n$  is defined by  $I_n = \int_0^1 e^x (2-x)^n dx$  for integers  $n \geq 0$ .

(i) Show that  $I_n = e - 2^n + nI_{n-1}$

2

|                  |                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 marks          | Result shown                                                                                                                                                                                                                                                                                                                                                                        |
| 1 mark           | Progress towards the required result                                                                                                                                                                                                                                                                                                                                                |
| Markers Comments | <p>a) Well done, however, students who incorrectly stated that <math>\frac{d}{dx}(2-x)^n = n(2-x)^{n-1}</math>, were either unable to obtain the appropriate Recurrence formula or attempted to <u>fudge</u> the result.</p> <p>(i) Errors when dealing with + and - signs were not uncommon.</p> <p>(ii) Students are again reminded of the importance of attention to detail.</p> |

### Sample solution

$$\begin{aligned}
 \text{Let } u &= (2-x)^n & v &= e^x & I_n &= [uv]_0^1 - \int_0^1 vu' dx \\
 u' &= -n(2-x)^{n-1} & v' &= e^x & &= \left[ e^x (2-x)^n \right]_0^1 - \int_0^1 -e^x n(2-x)^{n-1} dx \\
 & & & & &= e^1 (2-1)^n - e^0 (2-0)^n + n \int_0^1 e^x (2-x)^{n-1} dx \\
 & & & & &= e - 2^n + nI_{n-1}
 \end{aligned}$$

(ii) Hence or otherwise, evaluate  $\int_0^1 e^x (2-x)^4 dx$

3

|                  |                                                                                   |
|------------------|-----------------------------------------------------------------------------------|
| 3 marks          | Correct answer                                                                    |
| 2 marks          | Significant progress towards correct answer                                       |
| 1 marks          | Demonstrates knowledge of how to apply the recursive formula or equivalent merit. |
| Markers Comments |                                                                                   |

### Sample solution

Required to find  $I_4$

$$\begin{aligned}
 I_0 &= \int_0^1 e^x dx & I_1 &= e - 2^1 + 1I_0 & I_2 &= e - 2^2 + 2I_1 & I_3 &= e - 2^3 + 3I_2 \\
 &= \left[ e^x \right]_0^1 & &= e - 2 + e - 1 & &= e - 4 + 2(2e - 3) & &= e - 8 + 3(5e - 10) \\
 &= e - 1 & &= 2e - 3 & &= 5e - 10 & &= 16e - 38
 \end{aligned}$$

$$\begin{aligned}
 I_4 &= e - 2^4 + 4I_3 \\
 &= e - 16 + 4(16e - 38) \\
 &= 65e - 168
 \end{aligned}$$

- c) (i) Show that both  $\sin^2 x + \cos^4 x$  and  $\sin^4 x + \cos^2 x$  can each be expressed as

$$1 - \frac{1}{4} \sin^2 2x$$

2

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| <b>2 marks</b>  | Both results shown                                                |
| <b>1 mark</b>   | Shows one of the required results                                 |
| <b>Markers</b>  | (i) Most students were successful in proving the required result. |
| <b>Comments</b> |                                                                   |

### Sample solution

$$\begin{aligned}
 \sin^2 x + \cos^4 x &= \sin^2 x + \cos^2 x \cos^2 x & \sin^4 x + \cos^2 x &= \sin^2 x \sin^2 x + \cos^2 x \\
 &= \sin^2 x + \cos^2 x (1 - \sin^2 x) & &= \sin^2 x (1 - \cos^2 x) + \cos^2 x \\
 &= \sin^2 x + \cos^2 x - \cos^2 x \sin^2 x & &= \sin^2 x + \cos^2 x - \sin^2 x \cos^2 x \\
 &= 1 - (\sin x \cos x)^2 & &= 1 - (\sin x \cos x)^2 \\
 &= 1 - \left( \frac{1}{2} \sin 2x \right)^2 & &= 1 - \left( \frac{1}{2} \sin 2x \right)^2 \\
 &= 1 - \frac{1}{4} \sin^2 2x & &= 1 - \frac{1}{4} \sin^2 2x
 \end{aligned}$$

- (ii) Show that  $\frac{1}{\sin^2 x + \cos^4 x} + \frac{1}{\sin^4 x + \cos^2 x} = \frac{8}{4 - \sin^2 2x}$

1

|                 |                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1 mark</b>   | Result logically shown                                                                                                                                            |
| <b>Markers</b>  | (ii) Well done by almost all students. It was pleasing to see that those who were unable to show the result in (i) were able to show the required result in (ii). |
| <b>Comments</b> |                                                                                                                                                                   |

### Sample solution

$$\begin{aligned}
 LHS &= \frac{1}{\sin^2 x + \cos^4 x} + \frac{1}{\sin^4 x + \cos^2 x} \\
 &= \frac{1}{1 - \frac{1}{4} \sin^2 2x} + \frac{1}{1 - \frac{1}{4} \sin^2 2x} \\
 &= \frac{2}{1 - \frac{1}{4} \sin^2 2x} \\
 &= \frac{8}{4 - \sin^2 2x}
 \end{aligned}$$

(iii) Hence, or otherwise, show that  $2 \leq \frac{1}{\sin^2 x + \cos^4 x} + \frac{1}{\sin^4 x + \cos^2 x} \leq \frac{8}{3}$

2

|                         |                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2 marks</b>          | Result logically shown                                                                                                                                                                                                       |
| <b>1 mark</b>           | Progress towards the required result                                                                                                                                                                                         |
| <b>Markers Comments</b> | (iii) Using $-1 \leq \sin x \leq 1 \Rightarrow 0 \leq \sin^2 x \leq 1$ enabled to successfully obtain the minimum and maximum values for the denominator and hence the maximum and minimum values for $\frac{8}{4-\sin^2 x}$ |

### Sample solution

Consider the range of the known function  $4 - \sin^2 2x$

$$0 \leq \sin^2 2x \leq 1$$

$$3 \leq 4 - \sin^2 2x \leq 4$$

$$\frac{1}{4} \leq \frac{1}{4 - \sin^2 2x} \leq \frac{1}{3}$$

$$2 \leq \frac{8}{4 - \sin^2 2x} \leq \frac{8}{3}$$

Hence  $2 \leq \frac{1}{\sin^2 x + \cos^4 x} + \frac{1}{\sin^4 x + \cos^2 x} \leq \frac{8}{3}$

- d) Prove that  $a^3 + 3a^2 + 2a$ , where  $a$  is a positive whole number, is divisible by 6 by mathematical induction.

3

|                         |                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3 marks</b>          | Valid proof by mathematical induction                                                                                                                                                                                        |
| <b>2 marks</b>          | Significant progress towards valid proof                                                                                                                                                                                     |
| <b>1 mark</b>           | Verification of the case for $n = 1$                                                                                                                                                                                         |
| <b>Markers Comments</b> | (iii) Using $-1 \leq \sin x \leq 1 \Rightarrow 0 \leq \sin^2 x \leq 1$ enabled to successfully obtain the minimum and maximum values for the denominator and hence the maximum and minimum values for $\frac{8}{4-\sin^2 x}$ |

### Sample solution

For  $a = 1$ :  $1^3 + 3(1)^2 + 2(1) = 6$  which is divisible by six hence true for  $a = 1$

Assume true for  $a = k$ . I.e.  $k^3 + 3k^2 + 2k = 6M$ ,  $M \in \mathbb{Z}$ .

For  $a = k + 1$ , r.t.p  $(k + 1)^3 + 3(k + 1)^2 + 2(k + 1)$  is divisible by 6.

$$\begin{aligned}
(k+1)^3 + 3(k+1)^2 + 2(k+1) &= k^3 + 3k^2 + 3k + 1 + 3k^2 + 6k + 3 + 2k + 2 \\
&= k^3 + 3k^2 + 2k + 3k^2 + 9k + 6 \\
&= 6M + 3(k^2 + 3k + 2) \\
&= 6M + 3 \underbrace{(k+2)(k+1)}_{\substack{\text{consecutive integers} \\ \text{hence 2 must be a} \\ \text{factor of one of the} \\ \text{even integer}}} \\
&\quad \underbrace{\hspace{1.5cm}}_{\substack{\text{Both 3 and 2 are factors} \\ \text{of this term.}}}
\end{aligned}$$

$\therefore$  both of these terms are divisible by 6.

Hence true by the process of mathematical induction.

**End of Exam**